

TASK 1 REPORT

Title

Data Cleaning and Preprocessing of Titanic Dataset

Objective

The objective of this task is to clean and preprocess the Titanic dataset by handling missing values, removing duplicate records, converting categorical data into numerical form, and preparing the dataset for machine learning analysis.

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Dataset Description

The Titanic dataset contains information about passengers aboard the Titanic, including demographic details, ticket information, fare, cabin details, and survival status. The dataset is commonly used for data analysis and machine learning projects.

Methodology

Step 1: Data Loading

The dataset was loaded using the Pandas library and inspected to understand its structure and contents.

Step 2: Data Exploration

Basic information about the dataset was examined using:

- head()
- info()
- describe()
- shape

Step 3: Handling Missing Values

Missing values were identified and handled using appropriate techniques:

- Numerical columns were filled using mean or median values.
- Categorical columns were filled using mode values.
- Columns with excessive missing values were removed when necessary.

Step 4: Removing Duplicate Records

Duplicate rows were checked and removed to improve data quality.

Step 5: Encoding Categorical Variables

Categorical columns such as Sex and Embarked were converted into numerical values using encoding techniques.

Step 6: Data Validation

The cleaned dataset was verified to ensure no significant missing values or inconsistencies remained.

Results

The dataset was successfully cleaned and prepared for further analysis and machine learning tasks. Missing values were handled, categorical variables were encoded, and duplicate records were removed.

Observations

- Several columns contained missing values.
- Data preprocessing improved dataset quality.
- The cleaned dataset is suitable for predictive modeling.

Conclusion

Data cleaning and preprocessing are essential steps in the data science workflow. Through this task, the Titanic dataset was transformed into a clean and structured format suitable for analysis and machine learning applications.

Learning Outcomes

- Understanding dataset exploration techniques.
- Handling missing values effectively.
- Removing duplicate records.
- Encoding categorical features.
- Preparing data for machine learning models.